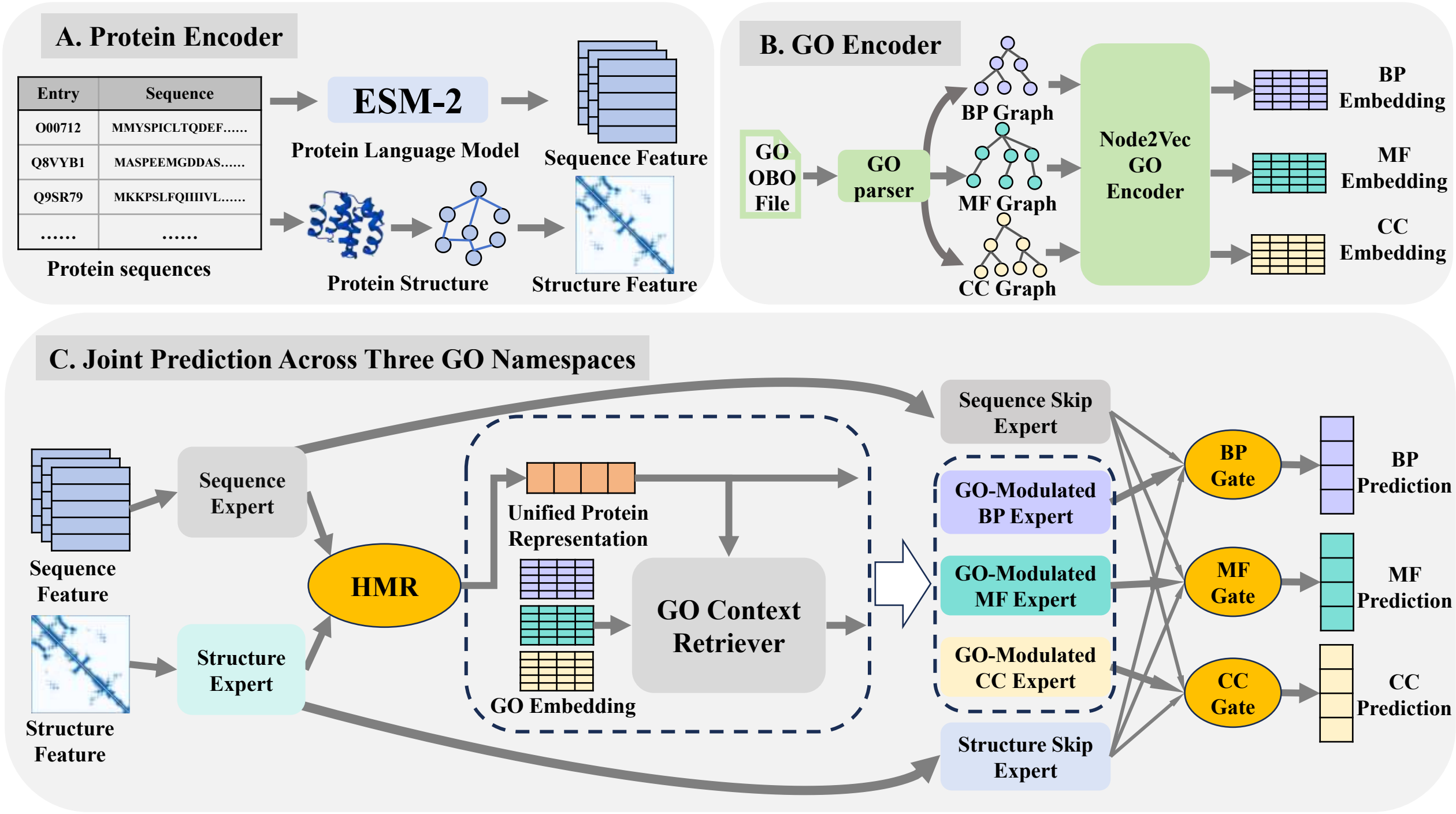

A. Protein Encoder
Entry
Sequence
O00712
MMYSPICLTQDEF.....
Q8VYB1
MASPEEMGDDAS.....
Q9SR79
MKKPSLFQIIIIVL.....
.....
.....
Protein sequences
ESM-2
Protein Language Model
Sequence Feature
Protein Structure
Structure Feature
B. GO Encoder
GO OBO File
GO parser
BP Graph
MF Graph
CC Graph
Node2Vec GO Encoder
BP Embedding
MF Embedding
CC Embedding
C. Joint Prediction Across Three GO Namespaces
Sequence Feature
Structure Feature
Sequence Expert
Structure Expert
HMR
Unified Protein Representation
GO Embedding
GO Context Retriever
Sequence Skip Expert
GO-Modulated BP Expert
GO-Modulated MF Expert
GO-Modulated CC Expert
Structure Skip Expert
BP Gate
MF Gate
CC Gate
BP Prediction
MF Prediction
CC Prediction